

BIOINFORMATICIAN (U.S. CITIZEN)*Persistence in decoding life using numbers and symbols*<https://www.linkedin.com/in/mousheng-xu-728bb34/>

SUMMARY: AI & Bioinformatics Engineer with 20+ years spanning bio-pharma, academia, and software. Strong Python/R/Java, statistical modeling, and large-scale pipeline automation; published nearly 20 peer-reviewed papers. Entrepreneurial and client-facing track record delivering data-driven solutions in real-world settings (NIH/NIDDK, DFCI/Harvard, biotech).

STATUS: U.S. citizen, available immediately.

HIGHLIGHTS OF EXPERTISE

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|---------------------------------------|----------------------------|-----------------------------|
| • <i>Certified</i> Statistics Modeler | • Biomarker Identification | • Cancer Dependency Map, |
| • GWAS, Geno-Pheno | • NGS, DNaseq, RNAseq, | cBioPortal, TCGA, Encode, |
| • PyTorch (IBM Certified) | ChIPseq, GUIDEseq, Super | HapMap, dbSNP, 1000G, |
| • Keras/Tensorflow | Enhancer Analysis | HGMD, OMIM |
| • Machine Learning | • Antibody Typing | • Cancer Biology, Oncology, |
| • NLP | • Mendelian Randomization | Genetic Epidemiology, |
| | | Population Genetics |

~ COMPUTER LANGUAGES & BIOINFORMATIC SKILLS ~

- R • JAVA • Perl • Python • C/C++ • HTML • JavaScript • React Native • XML • HAIL • AWS • Google Cloud
• Samtools • Hisat • Bowtie • STAR • edgeR • ROSE • Bcftools • GATK • Plink • ShapeIt • Eagle
• Minimac • SNPTest • RVTests • METAL • TwoSampleMR • GCTA-cojo • Wireguard • ChatGPT API

EDUCATION

Postdoc Fellow in Cancer Genetic Epidemiology, Harvard T.H. Chan School of Public Health
Predoc Fellow in Bioinformatics, Harvard Medical School
Ph.D. in Bioinformatics & Systems Biology, Boston University
Master of Science in Computer Science, University of Kansas
Master of Science in Biochemistry & Molecular Biology, University of Kansas Medical Center
Biostatistics Ph.D. Program, Ohio State University; Certified Statistics Modeler
Bachelor of Science in Genetics, Wuhan University, P. R. China

CAREER SUMMARY**Head of Biosciences | Sophyics Pte. Ltd. | Singapore**

2025.01– Present

Leading the scientific strategy and execution at a deep-tech startup exploring quantum-physics-based innovations with potential biomedical applications.

Head of Bioinformatics | D2M Biotherapeutics | Natick, MA

2021.07–2024.11

- Setup a secure intranet with VPN access to the company computing server using wireguard
- Assisted the **fundraising** initiatives through effective investor communication and engagement
- Enhanced the drug development process by delivering in-depth causal gene analysis results, thereby aiding in the creation of innovative pipelines

continued...

- **Developed & Integrated Colocalization and Medelian Randomization Analysis Pipeline:** created and seamlessly integrated a Colocalization Analysis pipeline with the Two Sample Mendelian Randomization (2sMR) pipeline. This optimized the identification of causal gene targets, significantly impacting disease research
 - Conversion Process Automation: Conceived and implemented an automated pipeline to transform GWAS summary statistics files into TwoSampleMR and GCTA-cojo compatible formats, enhancing efficiency
 - TwoSampleMR Automation: Designed an automated process to execute TwoSampleMR, bolstering both efficiency and accuracy in data handling
 - Feature Enhancement: Augmented pipeline functionality by introducing the ability to utilize SNP proxies for local outcome files, a capability not present in the standard TwoSampleMR package.
 - Comprehensive 2sMR Analysis: Completed an exhaustive 2sMR analysis on all available pQTL datasets in ieugwas, including selected pQTL datasets from peer-reviewed journals, against over 400+ cancer and immune diseases. This critical analysis played a vital role in pinpointing causal gene targets for drug development, marking a key milestone that successfully secured company fundraising
 - Genome-wide Colocalization Scan: Conducted a comprehensive colocalization scan of eQTL, pQTL, and sQTL versus hundreds of selected disease outcomes, and generated 11+ million results
- Established an Antibody Domain Identification (Antibody Typing) pipeline for PacBio DNaseq datasets using IgBlast, facilitating the design of antibodies for drug discovery
- Installed AlphaFold and built a pipeline to facilitate protein 3D structure analysis for drug discovery
- Conducted a comprehensive evaluation of **AWS** cloud computing capabilities, including a detailed price quote analysis and account setup to assess the potential of AWS to alleviate the computational bottlenecks encountered in the colocalization analysis

Sr. Data Analyst, Sr. Statistical Geneticist | Booz Allen Hamilton | Boston, MA 2018.03–2021.07

- As the chief architect, designed the overall architecture and implemented the initial version of a Natural Language Processing (NLP) pipeline for a medical data management system belonging to the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) to tag biological study data with Common Data Elements (CDEs)
 - Preprocessed the CDEs and study data to expected data formats
 - Further preprocessed the input data with letter case canonicalization, tags removal, punctuation stripping
 - Applied various information retrieval techniques including stop words removal, stemming, Latent Semantic Indexing (LSI), Matrix Similarity, and TF-IDF modeling
- Key player in the data analysis of 650k GWAS of the **Million Veteran Program**, a national research program to study how genes, lifestyle, and military exposures affect health and illness, the largest cohorts in the world with medical records
 - Led a group and developed a user-friendly pipeline to validate geno/pheno-typic data and achieved 100x speed improvement compared to existing ones.
 - Led the Mitochondria DNA project to QA/QC the microarray genotypic data with call rate, intentional duplicates, and called WGS variants, delivered the high-quality information to genomic researchers, and evaluated the feasibility of MT DNA imputation
 - In charge of the ancestry (or DNA, chromosome) painting project to infer local ancestry of all the MVP samples, analyzed the ancestry composition of MVP African, European, and Hispanic samples and showed the power of local ancestry inference (LAI) for medical and population genetic studies
 - Initiated, designed, and implemented the world's first algorithm to validate father-mother-child trioship using genomic information and statistical theory
 - Lead the study of popular WGS/WES variant calling pipelines based on GATK4, Docker, WDL, and cloud computing.

Bioinformatics Analyst II | Dana-Farber Cancer Institute | Harvard Medical School | Boston, MA 2016.05–2018.03

- Responsible for the data analyses of several medical oncology research projects using RNAseq, ChIPseq, and GUIDEseq, and co-authors of four publications
 - Validated the impact of a drug compound on the expression of the target gene and analyze the synthetic transcription elongation factors license transcription across repressive chromatin (*Science*)
 - Validated the biological mechanism for the YAP1-Mediated Suppression of USP31 that enhances NFκB activity to promote **Sarcomagenesis** through differential gene expression analysis and histone modification (H3K27Ac) enhancer analysis (*Cancer Res.*)
 - Confirmed the degradation of the target protein of degrader TRIM24 through statistical analysis of gene expression (*Nat Chem Biol*)

- Analyzed the targeting MYC dependency in **ovarian cancer** through inhibition of CDK7 and CDK12/13 (*Elife*)
- Identified the molecular mechanism of **drug tolerant** in **B cell lymphoma** tumor cells by super enhancer analysis using epigenetic information (histone modifications) and gene differential expression analysis to be the loss of the drug target gene BCL2 in drug tolerant tumor cells (*Cancer Cell*)
- As the sole bioinformatics analyst, established new and maintained existing analysis pipelines and tools to take care of all research needs
 - Implemented a multivariate RNAseq analysis pipeline with downstream analysis and report
 - Improved existing ChIPseq pipelines to study super enhancer and gene expression regulation
 - Modified GUIDEseq to identify Double Strand Breaks (DSB) introduced by radiation
 - Pipelined variant calling from RNAseq datasets
 - Integrated the Achilles shRNA knock-down dataset from the Cancer Dependency Map project into gene discovery analysis
 - AWS cloud computing for system data backup and cloud web hosting

Founder/CEO/CTO | HUBEI VIDAGEN GENE TECHNOLOGY CO. LTD. | Wuhan, China 2011.05–2015.12

Operated with end-to-end ownership (engineering, analytics, delivery), demonstrating forward-deployed execution in clinical and product settings.

- Scaled the company to 35 employees, fostering a collaborative and innovative work environment
- Established business relationship with dozens of hospitals and healthcare organizations
- Designed and directed the development of dozens of genetic testing products and obtained 10 patents
- Built a Chinese genome-wide phenotype prediction system with a proprietary geno-pheno database, a prediction algorithm, and computer programs
- Directed the research in collaboration with a local hospital to estimate warfarin dosage in Han Chinese patients using clinical attributes and genetic variants of drug metabolizing gene VKORC1 and CYP2C9 (*Pharmacogenomics 2015*)

Postdoc Research Fellow, Cancer Genetic Epidemiology

HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH | Boston, MA 2009.05–2011.05

- Publications: co-authored 7 papers – including 1 *Nature Human Behav.* Mostly cancer GWAS and cancer survival analysis. All work cited below are published.
- Genome wide association studies
 - Mole count and **melanoma** risk: first identified variants associated with melanocytic nevi with a two-stage GWAS, then verified the gene NID1 linked with these variants are associated with melanoma risk by gene expression analysis
 - Handedness: estimated handedness heritability with SNPs using EMMAX, identified a few variants close to the association significance level later verified by other studies, and accidentally discovered that Jews have a much higher percentage of left-handedness by principal component analysis (PCA). This work is later published in *Nature Human Behav.* after joining an international meta-analysis of 1.8 million samples
 - **Basal cell carcinoma** (BCC): first identified SNPs associated with BCC by GWAS, and then identified 4 biological pathways associated with BCC by pathway enrichment analysis
 - **Pancreatic adenocarcinoma** survival: identified a germline variant that is associated with survival in pancreatic adenocarcinoma patients by a two-stage GWAS analysis and Cox proportional hazard and Kaplan-Meier plot statistical methods with age, sex, and clinical stage
- Disease risk prediction using environmental and genetic risk factors
 - ABO blood type and **pancreatic cancer** risk: observed an increased risk in participants with A1 and A2 alleles and found secretor is not an effect modifier by logistic regression analysis
 - blood type and **breast cancer** risk and survival: performed association analysis and survival analysis between serologic or genetic ABO blood type and breast cancer risk and noted no association

Predoc Research Fellow | CHANNING LAB, HARVARD MEDICAL SCHOOL | Boston, MA 2006.05–2009.05

- Conducted Ph.D. thesis research on ‘Computational Approach for The Identification of Functional Regulatory Genetic Polymorphism’, using gene regulatory elements, DNA sequence features, and genetic markers
- Conducted leading-edge study on development of computational model to predict asthma exacerbation with thousands of SNPs, leveraging on **clinical attributes** (age, sex, pre-bronchodilator FEV1 percentage predicted, and treatment group), and 550k genetic markers, and **machine learning** (Random Forests)
 - Won Travel Award to 2009 American Thoracic Society (ATS) International Conference in San Diego, and presented research findings at ‘**Scientific Breakthroughs of the Year**’ special symposium
 - Publication in **BMC Med Genet** had been accessed thousands of times in 6 months

Ph.D. Boston University | Boston, MA

2004.09–2006.05

- Involved in the pilot phase of Encyclopedia of DNA Elements (ENCODE) project to identify all functional elements in human genome sequence by annotating histone modification epigenetic tracks: co-authored “The ENCODE Project Consortium. Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project” (*Nature*)

Sr. Software Engineer | GENE LOGIC INC. | Berkeley, CA

2001.09–2003.11

Gene Logic provides comprehensive genomic platforms and bioinformatics products.

- As the principal engineer in the middle tier group, designed, implemented, and successfully delivered **GeneExpress 2.0**, a microarray gene expression data management software product providing support for exploration and analysis of gene expression data for tens of thousands of biological samples, together with associated clinical data, gene annotation, and metabolic pathway information included in KEGG and BioCarta. GeneExpress is Gene Logic’s flagship product and is currently used by the top pharmaceutical companies.
- Strove for a higher software testing standard, pioneered and constructed a complete set of unit testing (**JUnit**) and functional testing.

Computational Scientist | Celltech R & D, Inc. | Seattle, WA

1998.08–2001.09

Celltech R&D Inc is a drug discovery and development company concentrated on new approaches to treat cancer, autoimmune disease, inflammation and bone disease.

- Expression Gene Array Database (EGAD) System - a Web based application system to register gene array data for drug discovery R & D (Tech Lead)
- Chemical Registration (CREG) System - a Web based application system to register and display chemical structures and related data for drug discovery R & D (Tech Lead)

AWARDS & HONORS

1st Class Prize in Math, 2nd Class Prize in Physics, 3rd Class Prize in Chemistry, University Competitions
 2nd Prize, Best Clinical Science Research Abstract,
 Massachusetts Thoracic Society (MTS) & MTS Research Allocations Committee, 2008;
 Travel Award, American Thoracic Society (ATS) International Conference, San Diego, 2009;
 Abstract Presentation at ‘Scientific Breakthroughs of the Year’ Special Symposium, ATS, 2009

INDEPENDENT SOFTWARE PROJECTS

Developed and published multiple mobile applications on the Apple App Store, showcasing full-stack development, design, and deployment skills.

- [TraitGen](#) (iOS, App Store) — a paternity test mobile app based on body traits which does not require DNA information for entertainment and educational purposes. *Built with React Native.*
- [Fast PSAT Words – Gold](#) (iOS, App Store) — a PSAT vocabulary learning app designed to improve SAT/PSAT verbal scores. *Developed in Java using Codename One.*
- [Fast SAT Words – Gold](#) (iOS, App Store) — a SAT vocabulary learning app designed to improve SAT/PSAT verbal scores. *Developed in Java using Codename One.*

SOCIAL ACTIVITY

Nominating Chair | Newton Lower Falls Improvement, a Non-Profit organization | Newton, MA

PUBLICATIONS

Published nearly 20 peer-reviewed papers, including in *Science*, *Nature*, *Cancer Research*, and *Pharmacogenomics*. Full list below.

1. <https://www.kaggle.com/moushengxu/keras-lstm-with-multi-d-in-output-for-stock-price> a tutorial on applying the **deep learning** algorithm **LSTM** to predict daily stock price changes using the Tensorflow/Keras framework (2018)
2. Cuellar-Partida G, etc. **Genome**-wide association study identifies 48 common genetic variants associated with handedness. **Nat Hum Behav**. 2020 Sep 28
3. Zeng M, Kwiatkowski NP, Zhang T, Nabet B, **Xu M**, Liang Y, Quan C, Wang J, Hao M, Palakurthi S, Zhou S, Zeng Q, Kirschmeier PT, Meghani K, Leggett AL, Qi J, Shapiro GI, Liu JF, Matulonis UA, Lin CY, Konstantinopoulos PA, Gray NS. Targeting MYC dependency in ovarian cancer through inhibition of CDK7 and CDK12/13. *Elife*. 2018 Nov 13;7:e39030. doi: 10.7554/eLife.39030.
4. Gechijian LN, Buckley DL, Lawlor MA, Reyes JM, Paulk J, Ott CJ, Winter GE, Erb MA, Scott TG, **Xu M**, Seo HS, Dhe-Paganon S, Kwiatkowski NP, Perry JA, Qi J, Gray NS, Bradner JE. Functional TRIM24 degrader via conjugation of ineffectual bromodomain and VHL ligands. **Nat Chem Biol**. 2018 Apr;14(4):405-412. doi: 10.1038/s41589-018-0010-y. Epub 2018 Mar 5.
5. Ye S, Lawlor MA, Rivera-Reyes A, Egolf S, Chor S, Pak K, Ciottie GE, Lee AC, Marino GE, Shah J, Niedzwicki D, Weber K, Park PMC, Alam MZ, Grazioli A, Haldar M, Xu M, Perry JA, Qi J, Eisinger-Mathason TSK. YAP1-Mediated Suppression of USP31 Enhances NFκB Activity to Promote Sarcomagenesis. **Cancer Res**. 2018 May 15;78(10):2705-2720.
6. Erwin GS, Grieshop MP, Ali A, Qi J, Lawlor M, Kumar D, Ahmad I, McNally A, Teider N, Worringer K, Sivasankaran R, Syed DN, Eguchi A, Ashraf M, Jeffery J, **Xu M**, Park PMC, Mukhtar H, Srivastava AK, Faruq M, Bradner JE, Ansari AZ. Synthetic transcription elongation factors license transcription across repressive chromatin. **Science**. 2017 Dec 22.
7. Peng Q, Huang S, Chen X, Yuan Y, Yu Y, Tao L, Zhang Z, **Xu M**. [Validation of warfarin pharmacogenetic algorithms in 586 Han Chinese patients](#). **Pharmacogenomics**. 2015;16(13):1465-74.
8. Wu C, Kraft P, Stolzenberg-Solomon R, Steplowski E, Brotzman M, **Xu M**, et al. Genome-wide association study of survival in patients with pancreatic adenocarcinoma. **Gut**. 2012 ,Nov 24.
9. The ENCODE Project Consortium. Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project. **Nature** 2007(447):799-816.
10. Zhang M, Liang L, **Xu M**, Qureshi AA, Han J. Pathway analysis for genome-wide association study of basal cell carcinoma of the skin. **PLoS One**. 2011;6(7):e22760. Epub 2011 Jul 28.
11. **Xu M**, Tantisira KG, Wu A, Litonjua A, Chu J, Himes BE, Murphy A, Weiss ST. **Genome Wide Association Study to Predict Severe Asthma Exacerbations in Children using Random Forests Classifiers**. **BMC Med Genet**. 2011 Jun 30;12:90.
12. Nan H, **Xu M**, Kraft P, Qureshi AA, Chen C, Guo Q, Hu FB, Curhan G, Amos CI, Wang LE, Lee JE, Wei Q, Hunter DJ, Han J. Genome-wide association study identifies novel alleles associated with risk of cutaneous basal cell carcinoma and squamous cell carcinoma. **Hum Mol Genet**. 2011 Jul 6.
13. Gates MA, **Xu M**, Chen WY, Kraft P, Hankinson SE, Wolpin BM. ABO blood group and breast cancer incidence and survival. **Int J Cancer**. 2011 Jun 1. doi: 10.1002/ijc.26220.
14. Nan H, **Xu M**, Zhang J, Zhang M, Kraft P, Qureshi AA, Chen C, Guo Q, Hu FB, Rimm EB, Curhan G, Song Y, Amos CI, Wang LE, Lee JE, Wei Q, Hunter DJ, Han J. Genome-wide association study identifies nidogen 1 (NID1) as a susceptibility locus to cutaneous nevi and melanoma risk. **Hum Mol Genet**. 2011 Apr 9. (Co-first author)
15. Wolpin BM, Kraft PL, **Xu M**, Steplowski E, Olsson ML, Arslan AA, Bueno-de-Mesquita HB, Myron G, Helzlsouer K, Jacobs EJ, Lacroix A, Gloria P, Stolzenberg-Solomon RZ, Zheng W, Albanes D, Allen NE, Amundadottir L, Austin MA, Boutron-Ruault MC, Buring JE, Canzian F, Chanock SJ, Gaziano JM, Giovannucci E, Hallmans G, Hankinson SE, Hoover RN, Hunter DJ, Hutchinson A, Jacobs KB, Kooperberg C, Mendelsohn J, Michaud DS, Overvad K, Patel AV, Sanchez MJ, Sansbury LB, Shu XO, Slimani N, Tobias GS, Trichopoulos D, Vineis P, Visvanathan K, Virtamo J, Wactawski-Wende J, Watters JL, Yu K, Zeleniuch-Jacquotte A, Hartge P, Fuchs CS. Variant ABO Blood Group Alleles, Secretor Status and Risk of Pancreatic Cancer: Results from the Pancreatic Cancer Cohort Consortium. **Cancer Epidemiol Biomarkers Prev**. 2010 Oct 22.
16. Murphy A, Chu JH, **Xu M**, Carey VJ, Lazarus R, Liu A, Szeffler SJ, Strunk R, Demuth K, Castro M, Hansel NN, Diette GB, Vonakis BM, Franklin Adkinson N Jr, Klanderman BJ, Senter-Sylvia J, Ziniti J, Lange C, Pastinen T, Raby BA. Mapping of numerous disease-associated expression polymorphisms in primary peripheral blood CD4+ lymphocytes. **Hum Mol Genet**. 2010 Oct.
17. Nadeau K, Montermini L, Mandeville I, **Xu M**, Weiss ST, Sweezey NB, Kaplan F. Modulation of Lgl1 by steroid, retinoic acid and Vitamin D models complex transcriptional regulation during alveolarization. **Pediatr Res**. 2010 Apr;67(4):375-81.
18. Degnan JH, Lasky-Su J, Raby BA, **Xu M**, Molony C, Schadt EE, Lange C. Genomics and genome-wide association studies: an integrative approach to expression QTL mapping. **Genomics**. 2008(92):129-33.
19. **Xu M** and Gauch S. Associate Biological Information Retrieval From Distributed Databases. **In Proceedings of the 1998 ACM CIKM 7th International Conference on Information and Knowledge Management**, pages 193-200, Bethesda, Maryland, November 1998.